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Palaeomagnetic results from the Lancer 1 stratigraphic drillhole, Officer Basin, Western Australia, and implications for Rodinia reconstructions

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A partial record of the positions of Australia during Middle to Late Neoproterozoic time is provided by palaeomagnetic results for samples from the Lancer 1 stratigraphic drillhole in Western Australia. Lancer 1 was drilled vertically to 1501 m, through essentially horizontal Neoproterozoic strata of the western Officer Basin. We studied 123 samples from 28 intervals of drillcore which were oriented by matching features (fractures, cross-beds, etc.) in the core with oriented acoustic scanner images of the drillhole walls. Three new palaeopoles are reported for red mudstones and sandstones (redbeds) of the Browne (44.5°N, 141.7°E, dp = 5.1°, dm = 9.0°), Hussar (62.2°N, 85.8°E, dp = 7.3°, dm = 14.6°), and Kanpa (74.0°N, 128.8°E, dp = 7.7°, dm = 14.8°) Formations of the *ca* 830–720 Ma Buldya Group (Supersequence 1), which exhibit stable, two-polarity magnetisations carried by fine-grained hematite and magnetite. The overlying *ca* 610–590 Ma Wahlgu Formation glaciogenic diamictite (Supersequence 3) yielded dispersed directions and an imprecise palaeopole that overlaps results from the glaciogenic Elatina Formation and other Late Neoproterozoic rock units. The results help to elaborate the Middle to Late Neoproterozoic apparent polar wander path for Australia and indicate, in agreement with palaeoclimatic data and previous palaeomagnetic studies, that the continent was slow-moving and occupied low latitudes at this time. Assuming that Australia and Laurentia were still joined at *ca* 780 Ma, comparison of the new Hussar Formation palaeopole with coeval Laurentian data favours AUSMEX, rather than SWEAT or AUSWUS, as the most likely configuration of these two continents in Rodinia. This preliminary study of Lancer 1 demonstrates the utility of acoustic scanner logs for orienting drillcores, as well as the scope for additional sampling and palaeomagnetic studies of Lancer 1, and other oriented drillcores, to yield a more continuous record of Australia's past motions and to provide magnetostratigraphic data for enhancing inter-basin correlations.

KEY WORDS: acoustic scanner, drillhole, Neoproterozoic, Officer Basin, palaeomagnetism, redbeds, Western Australia.

INTRODUCTION

The Neoproterozoic history of the Australian craton is closely associated with that of the Rodinia supercontinent (Hoffman 1991; Dalziel 1997; Weil *et al*. 1998; Pisarevsky *et al*. 2003b). However, owing to the paucity of reliable (well-dated with proven or strongly suggested primary remanence) Neoproterozoic palaeomagnetic data for Australia, a variety of reconstructions of Australia as part of Rodinia have been proposed, including SWEAT (Dalziel 1991; Moores 1991), AUSWUS (Brookfield 1993; Karlstrom *et al*. 1999; Burrett & Berry 2000), and AUSMEX (Wingate *et al*. 2002). Of these, only the AUSMEX model is supported by a reliable palaeopole from the 1070 Ma Bangemall sills (Wingate *et al*. 2002). There are no reliable data between this and the next youngest reliable pole, from the 755 Ma Mundine Well dykes (Wingate & Giddings 2000), which probably represents a time during or after the breakup of Rodinia. Neoproterozoic palaeomagnetic data are insufficient to verify proposed reconstructions of Australia both during and after Rodinia time.

The Precambrian palaeomagnetic record is fragmentary, in part because most palaeomagnetic results have been obtained from intrusions or lava flows that cooled relatively quickly, or from limited outcrops of

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sedimentary strata, and so represent relatively short time intervals. In the absence of well-exposed Neoproterozoic successions, detailed sampling of sedimentary strata in deep continuous diamond drillcores can provide a more complete record of the geomagnetic field, and the samples are not subject to remagnetisations caused by weathering or lightning strikes. The inability to orient most drillcores accurately, however, has previously made them unpopular subjects for palaeomagnetic study, because non-oriented core samples yield only inclination (palaeolatitude) information (Pisarevsky *et al.* 2001). A method of core orientation based on isolating a recent viscous overprint (VRM) aligned with the present Earth's field (Cottrell & Tarduno 2003) is not applicable in many drillcores owing to the presence of a so-called steep drilling remanence with stability similar to, or higher than, the viscous one (Pisarevsky *et al.* 2001).

Pisarevsky *et al.* (2001) reported a palaeomagnetic study of core samples from the Empress 1A deep drillhole in the Officer Basin of Western Australia. In that study, the core samples were not oriented in azimuth, and only palaeomagnetic inclinations could be determined reliably. These data show that Australia occupied low latitudes during most of the Neoproterozoic, but provide no constraints on the orientation of the continent and so are of limited use in assessing reconstructions of Australia with other continents.

In the current study, we employed information from acoustic scanner logs to orient intervals of drillcore from another deep stratigraphic drillhole, Lancer 1, located 260 km northwest of Empress 1A, in the western Officer Basin (Figure 1). To the best of our knowledge, this is the first use of acoustic scanner logs to orient drillcore in palaeomagnetic studies of Precambrian rocks in Australia. In this study, we: (i) assess the usefulness and reliability of acoustic scanner images to orient diamond drillcores; (ii) conduct palaeomagnetic and rock-magnetic measurements of selected oriented intervals of drillcore, and determine which parts of the core are most suitable for more detailed study; and (iii) obtain four new palaeopoles that provide much-needed new data points on the sparsely populated apparent polar wander path (APWP) for Neoproterozoic Australia.

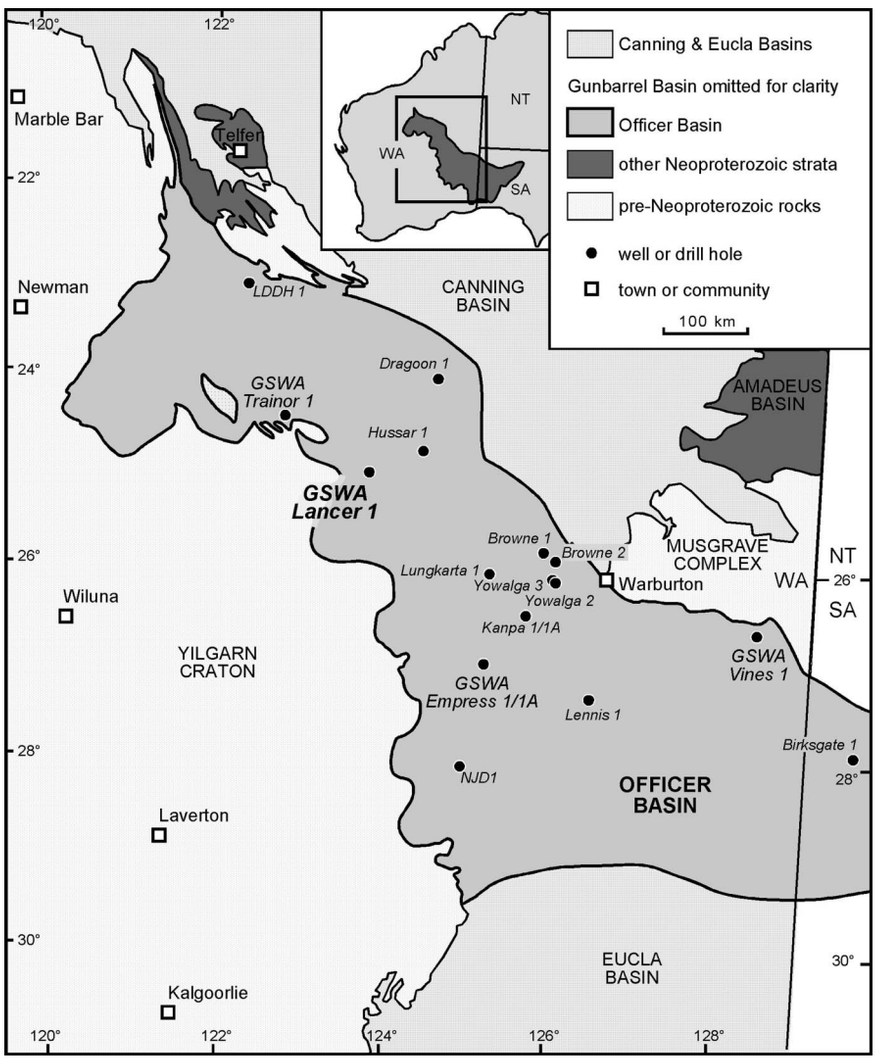
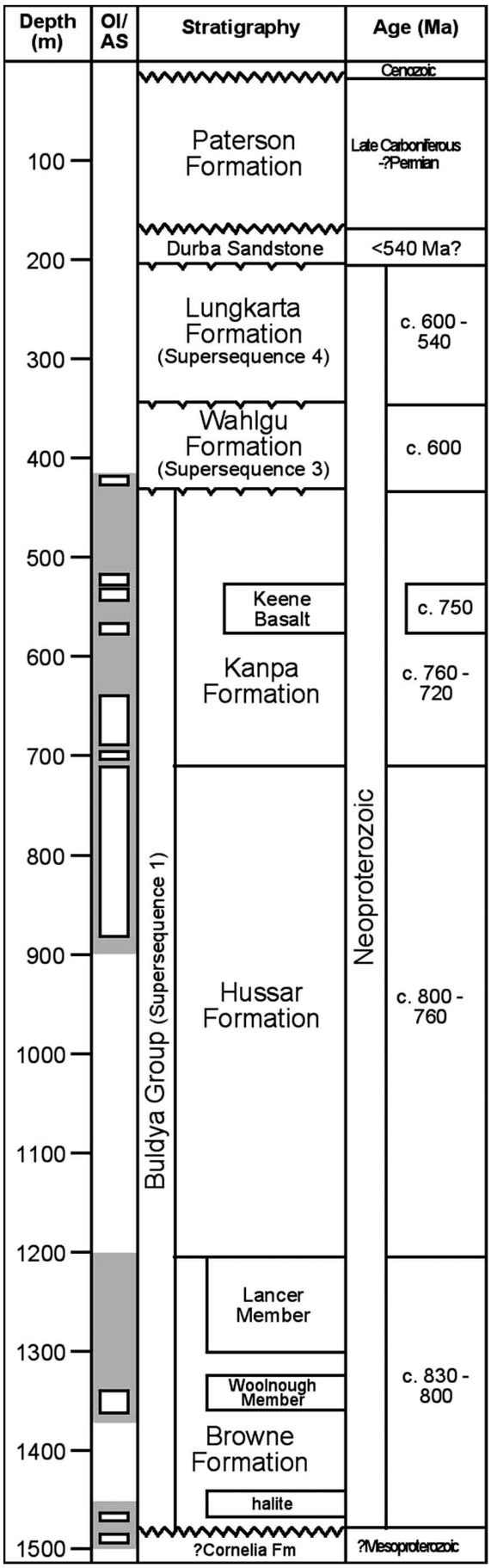


Figure 1 Simplified geology of part of Western Australia, showing the locations of Lancer 1 and other drillholes in the Officer Basin.

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GEOLOGICAL CONTEXT

The Officer Basin contains a thick, though stratigraphically discontinuous, Neoproterozoic record ranging in age from *ca* 850 to 543 Ma. It is inferred to have been part of the larger Centralian Superbasin (Walter *et al.* 1995), also incorporating the Amadeus, Ngalia, Georgina and other basins, during the Neoproterozoic. Each basin has a broadly similar Neoproterozoic stratigraphy, subdivided into four supersequences.

Lancer 1 was drilled in 2003 by the Geological Survey of Western Australia (GSWA) in the western Officer Basin (Figure 1) and continuously diamond cored from 104 m to a total depth of 1501.3 m. Lancer 1 is located at 25°02′44.5″S, 123°45′20.1″E (GDA94), 390 km east-northeast of Wiluna, in the western platform structural zone (Haines *et al.* 2004; Mory & Haines 2005; Simeonova & Iasky 2005), where relatively undeformed Officer Basin strata overlie Mesoproterozoic and older rocks, and are themselves overlain by Palaeozoic strata of the Gunbarrel Basin. Below Cenozoic strata and the Upper Carboniferous to Permian Paterson Formation of the Gunbarrel Basin, Lancer 1 penetrated the Neoproterozoic to Palaeozoic Durba Sandstone, and the Neoproterozoic Lungkarta, Wahlgu, Kanpa, Hussar and Browne Formations of the Officer Basin (Figure 2). Based on regional Centralian Superbasin stratigraphy the Lungkarta Formation is tentatively assigned to Supersequence 4, the Wahlgu Formation to Supersequence 3, and the remainder to Supersequence 1 (Haines *et al.* 2004; Grey *et al.* 2005). If Supersequence 2 was ever deposited in this area, it was eroded prior to deposition of Supersequence 3. The poorly age-constrained Durba Sandstone could belong to either the Officer or Gunbarrel Basin. In this study, we report results for the Wahlgu, Kanpa, Hussar and Browne Formations. The Middle to Late Neoproterozoic age of these strata is reasonably well established by a variety of data, including detrital-zircon geochronology, palynology, stromatolite biostratigraphy, preliminary palaeomagnetic studies, limited isotope chemostratigraphy, and lithostratigraphic correlation with dated successions in other areas (Carlsen & Grey 1998; Stevens & Adamides 1998; Pisarevsky 2001; Apak *et al.* 2002; Haines *et al.* 2004; Grey *et al.* 2005).

In addition, two samples were collected at the bottom of the core from well-cemented, medium-grained, cross-bedded sandstone tentatively assigned to the Cornelia Sandstone and presumed to be of Mesoproterozoic age. The contact with the overlying Browne Formation

Figure 2 Stratigraphy of the Lancer 1 drillhole (after Haines *et al.* 2004; Mory & Haines 2005), showing the intervals over which the acoustic scanner (AS) log was acquired (shaded sections) and the intervals (OI, white rectangles) over which oriented samples were taken. Officer Basin age constraints are based on interpretations by Grey *et al.* (2005). Supersequence 2, as recognised in the eastern Officer and Amadeus Basins, is not present in Lancer 1 due to non-deposition or erosion.

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is erosional, and a hiatus of uncertain duration is indicated.

The Browne Formation is the basal unit of the Officer Basin at this locality and, together with the Hussar and Kanpa Formations, is part of the Buldya Group. The age of the Buldya Group is considered to range from about 850 Ma to about 720 Ma, with the age of the Browne Formation about 830–800 Ma (Grey *et al.* 2005). In Lancer 1, the Browne Formation consists of a lower mudstone and evaporite succession, a middle stromatolitic dolostone and mudstone unit named the Woolnough Member, and an upper sandstone-dominated unit named the Lancer Member (Figure 2). The average structural dip in the Browne Formation is about 2°.

The Hussar Formation overlies the Browne Formation with a sharp contact that may represent a minor erosional hiatus. The age of the Hussar Formation is interpreted to be 800–760 Ma, based on its correlation with the Burra Group of South Australia (Grey *et al.* 2005 and references therein). The Boucaut Volcanics, which are at or below the base of the Burra Group, have a U–Pb SHRIMP zircon age of 777 ± 7 Ma (Preiss 2000). The lower half of the Hussar Formation consists predominantly of mudstone, with minor interbedded sandstone, dolostone and evaporite, whereas the upper part of the unit is predominantly sandstone, with minor interbedded mudstone and dolostone. Beds are horizontal, sometimes with high- to low-angle cross-bedding; some intervals are massive or structureless.

The Kanpa Formation overlies the Hussar Formation with a sharp conformable contact and forms the upper unit of the Buldya Group in the Lancer 1 drillhole. The lower Kanpa Formation consists mainly of mudstone, claystone and sandstone, and the upper part of stromatolitic dolostone, sandstone and mudstone. The Kanpa Formation also contains grey shale beds with carbonaceous disks tentatively identified as megascopic alga *Chuaria* sp. (Haines *et al.* 2004), possible ash beds, and a 49 m-thick basalt unit, the Keene Basalt. The basalt consists of four or possibly five flows with tops and bases marked by fine-grained amygdaloidal basalt and flow centres of coarse-grained massive basalt. The age of the Kanpa Formation is estimated to be 760–720 Ma, constrained by stromatolite biostratigraphy, palynology, isotope chemostratigraphic correlation with the 758 Ma Ombombo Supergroup in Namibia, and detrital-zircon age constraints (Hoffman *et al.* 1996; Grey *et al.* 2005).

The Wahlgu Formation overlies the Kanpa Formation with an erosional disconformity, although the Kanpa Formation and all overlying units are virtually horizontal. The Wahlgu Formation is constrained to be younger than *ca* 720 Ma and older than the Early Palaeozoic Table Hill Volcanics, which lie at a higher stratigraphic level in Empress 1A (Stevens & Apak 1999). The Wahlgu Formation consists predominantly of glacial diamictite, but the basal part of the unit includes well-sorted sandstone, mudstone and conglomerate. If correlation of this unit with the Marinoan glacial succession elsewhere in Australia (Walter *et al.* 1995; Haines *et al.* 2004; Grey *et al.* 2005) is correct, then it has an age of about 600 Ma.

SAMPLING AND ANALYTICAL METHODS

A wireline acoustic scanner log was acquired over three intervals of the vertical drillhole: from total depth (1501 m) to 1446 m, 1368–1198 m, and 902–425 m (Figure 2), providing an oriented acoustic image of the wall of the drillhole for the Mesoproterozoic strata, parts of the Buldya Group, and the basal Wahlgu Formation. We collected samples for palaeomagnetic study from oriented intervals for which azimuthal orientation could be determined using the acoustic scanner composite log (available on CD accompanying Haines *et al.* 2004 and Mory & Haines 2005). Once an individual piece of core was oriented, the oriented interval was extended by piecing together the core above and below it as far as possible.

The acoustic scanner is a wireline tool that is oriented in azimuth by magnetic compass and measures drillhole trajectory by gyroscope. The tool sends a focused sonic pulse as it rapidly spins about the drillhole axis to give complete coverage of the drillhole wall as it is being raised in the hole. The tool measures both the time that the pulse takes to be reflected back to the receiver (transit time), which gives a very accurate picture of the shape of the drillhole wall, and the strength of the return pulse (acoustic amplitude), which gives an indication of the relative hardness of the part of the wall being sampled. The latter values are

Table 1 Palaeomagnetic results from Lancer 1 samples.

Formation	Age (Ma)	*N*	*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	Plong (°E)	dp,dm (°)	λ_p (°)	Polarity
Wahlgu	*ca* 600	13	40	216.0	−14.5	3	31.5	42.3	175.7	16.5,32.3	−7.4	Mixed
Keene	*ca* 750	12	30	–	32.4	–	–	–	–	–	17.6	Reverse
Kanpa	760–720	6	11	181.4	17.7	23	14.3	74.0	128.8	7.7,14.8	9.1	Reverse
Hussar	800–760	17	29	163.3	4.6	7	14.6	62.2	85.8	7.3,14.6	2.3	Mixed
Browne	830–800	14	27	193.4	−33.8	27	7.9	44.5	141.7	5.1,9.0	−18.5	Reverse
?Cornelia	>1000	2	6	219.6	45.5	42	10.5	54.5	225.7	8.5,13.4	27.0	Reverse

N, *n*, number of samples, number of specimens; *D*, *I*, declination, inclination; *k*, Fisher's (1953) precision parameter; α_{95}, semi-angle of 95% circle of confidence about mean direction; Plat, Plong, latitude, longitude of palaeopole; dp, dm, semi-axes of cone of 95% confidence about mean pole; λ_p, palaeolatitude; Polarity, north declination is assumed to be normal polarity.

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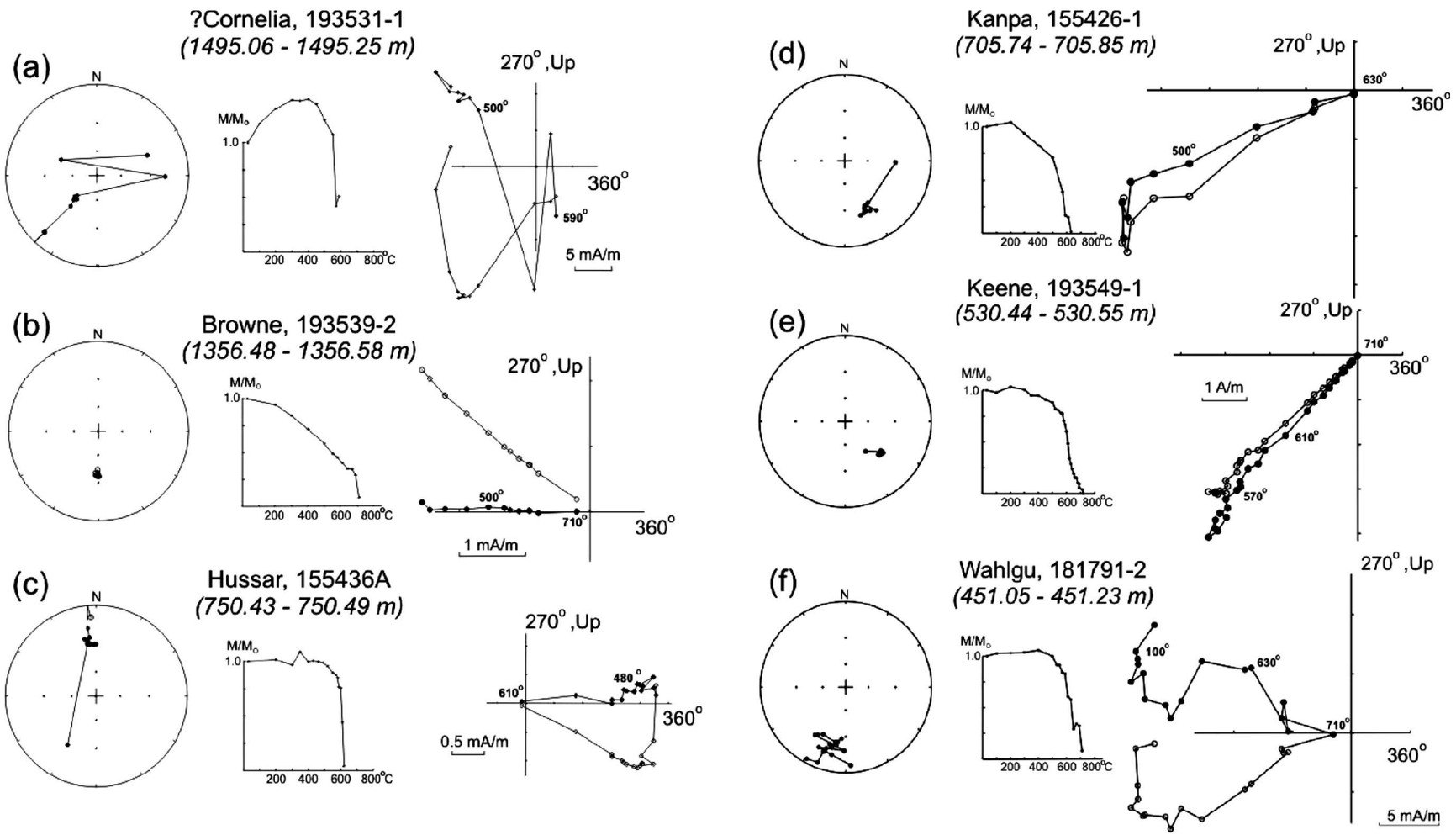


Figure 3 Examples of thermal demagnetisation of specimens from Lancer 1. Orthogonal plots (in geographical coordinates) show trajectories of magnetisation vector end-points during demagnetisation; open (closed) symbols refer to the vertical (horizontal) plane. Curves show changes in remanence intensity during demagnetisation. Stereoplots show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols.

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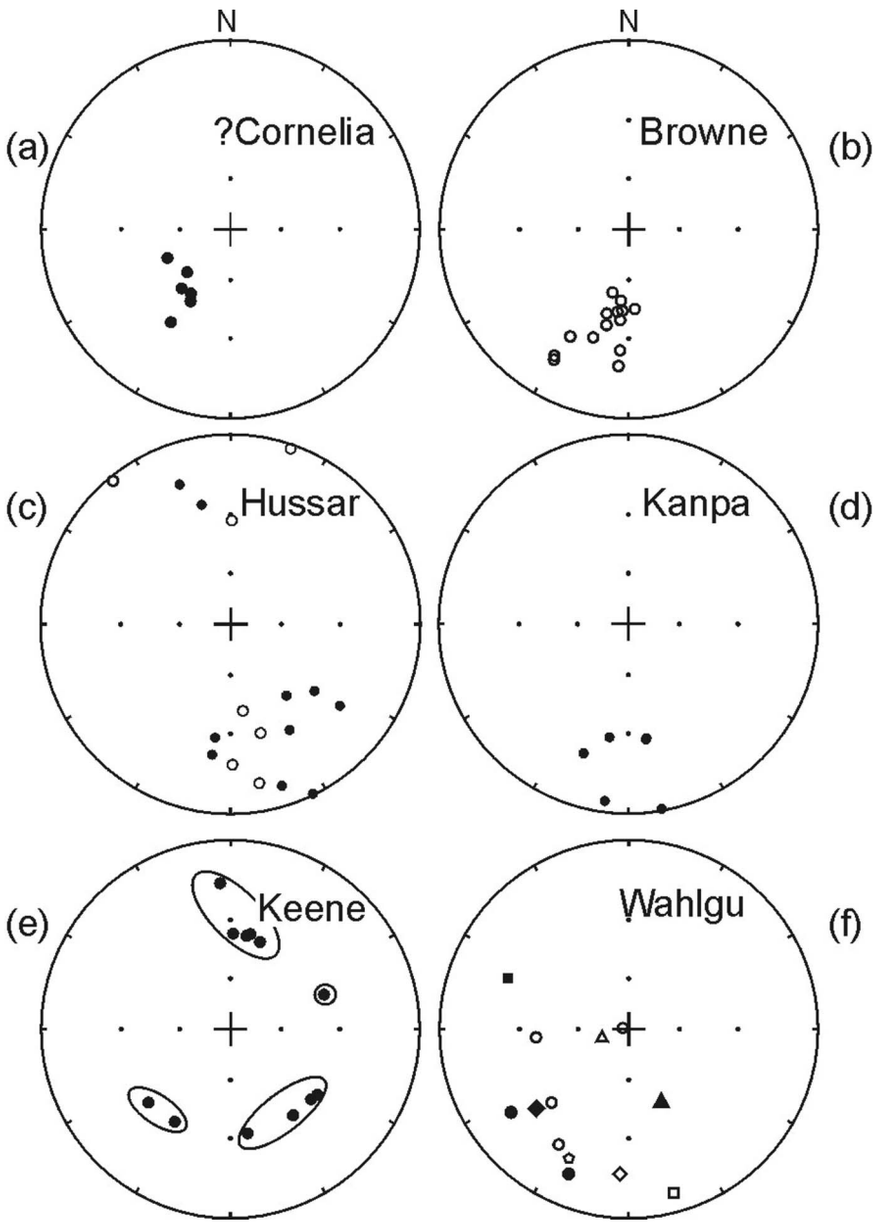


Figure 4 Stereoplots (stereographic projection) for parts of the Lancer 1 drillcore. Upwards (downwards) pointing palaeomagnetic directions are shown with open (closed) symbols. In Figure 4f, various symbols are used to show different oriented intervals.

processed to give an oriented three-dimensional acoustic image of the drillhole wall which can be compared to the diamond drillcore. The pixel size of the image generated by the tool is dependant on the size of the drillhole and, in the case of Lancer 1, is slightly smaller than 1 cm^2. An error in azimuthal orientation is likely to be in the order of 10° (except where the rocks are highly magnetic), and in vertical orientation by less than 5°.

A total of one hundred and twenty-three 5 to 7 cm-diameter oriented core pieces were obtained from 28 oriented intervals (Figure 2). A 25 mm-diameter palaeomagnetic sample was drilled along the central axis of each piece, and cut into standard-size specimens (22 mm high) for palaeomagnetic analysis. Remanence composition was determined by detailed stepwise thermal demagnetisation ($\leq$20 steps, to $>$700°C), using a Magnetic Measurements thermal demagnetiser and the 2G-755R cryogenic magnetometer. Stepwise alternating field (AF) demagnetisation ($\leq$26 steps, up to 120 mT) was also applied to igneous rocks of the Keene Basalt, using the 2G-600 automated degaussing system. AF demagnetisation was not effective for the sedimentary rocks, in which the main remanence carrier is hematite. To monitor possible mineralogical changes during heating, magnetic susceptibility was measured in selected samples after each heating step using a Bartington MS2 susceptibility meter. Magnetic mineralogy was investigated from demagnetisation characteristics and, in selected samples, from detailed variation of susceptibility *vs* temperature (20–700°C) obtained using the Bartington meter in conjunction with an automated Bartington furnace. These latter studies were limited to the Keene Basalt, owing to the low magnetic susceptibilities of the sedimentary rocks ($0.1-10\times10^{-5}$ SI units). Magnetisation vectors were isolated using principal-component analysis (Kirschvink 1980). All vectors were defined with

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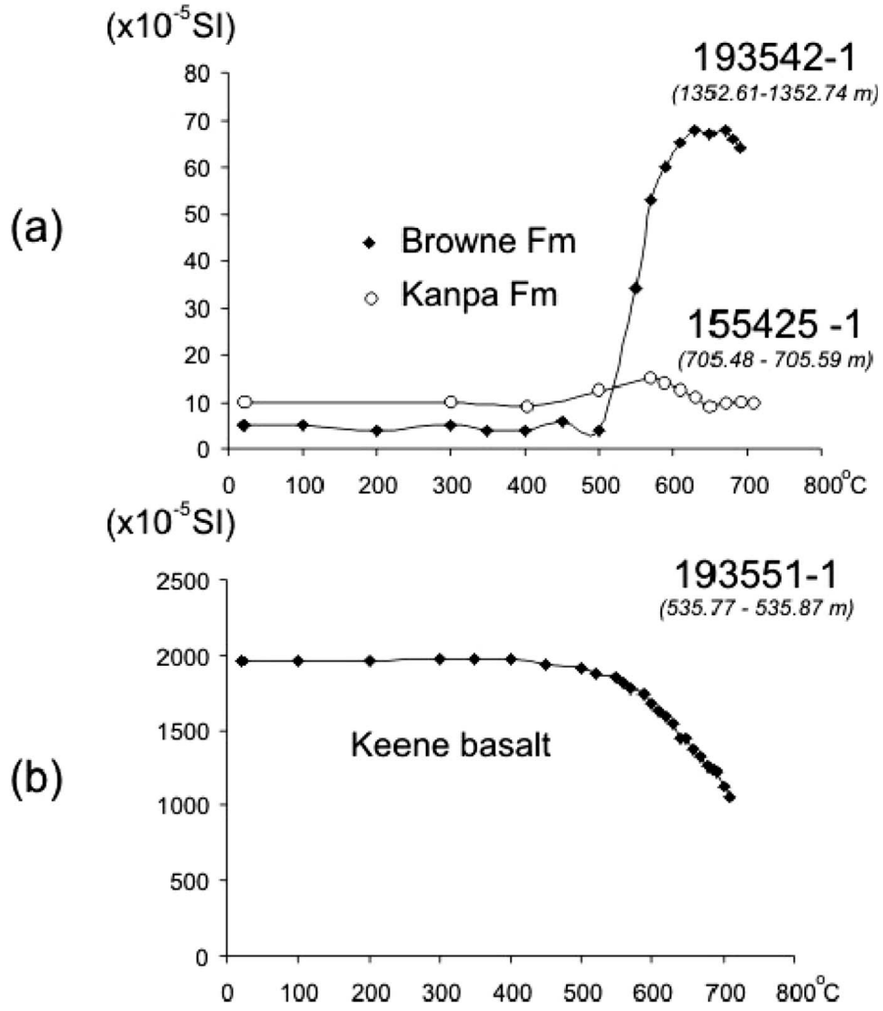


Figure 5 Magnetic susceptibility *vs* temperature curves for Kanpa and Browne Formations and Keene Basalt. Measurements of susceptibility were made stepwise after cooling samples to room temperature.

a minimum of four data points and a maximum angular deviation (MAD) of 10°. The results from each formation are described below and summarised in Table 1.

RESULTS

?Cornelia Sandstone

Two samples of grey sandstone were collected from a single oriented interval (1494.48–1495.25 m). A stable remanence isolated by thermal demagnetisation between 200 and 500°C differs significantly from that observed in samples of the overlying Browne Formation. Heating above 500°C causes changes in magnetic minerals, including production of magnetite from non-magnetic minerals such as pyrite or siderite and associated random directions (Figure 3a). Because only two samples (three specimens each) were measured, the statistics are calculated at specimen level (Table 1; Figure 4a).

Browne Formation

Fourteen samples of red mudstone, siltstone and sandstone were collected from two oriented intervals (1342.25–1358.49 m and 1469.95–1474.35 m) with each interval having several independent orientation features. Thermal demagnetisation revealed a stable, high-temperature, single-polarity remanence in all samples (Figure 3b). Magnetic susceptibility increases above 500°C (Figure 5a), suggesting generation of magnetite by breakdown of non-magnetic minerals. However, in this case, there was no change in the direction of magnetisation (Figure 3b). Similar susceptibility characteristics were reported for Browne Formation redbeds in the Empress 1A drillcore (Pisarevsky *et al*. 2001). Remanence directions are highly consistent within and between samples, and between oriented intervals (Table 1; Figure 4b).

Hussar Formation

Seventy-two samples of pink and red sandstone and mudstone were collected from 25 oriented intervals between 710.80 and 885.20 m. Unfortunately, most sandstone samples are magnetically unstable, and only 17 samples (mostly mudstones) yielded a stable remanence carried by magnetite and hematite (Figure 3c). Two polarities are present, with directions oriented upwards to the north and downwards to the south (Table 1; Figure 4c).

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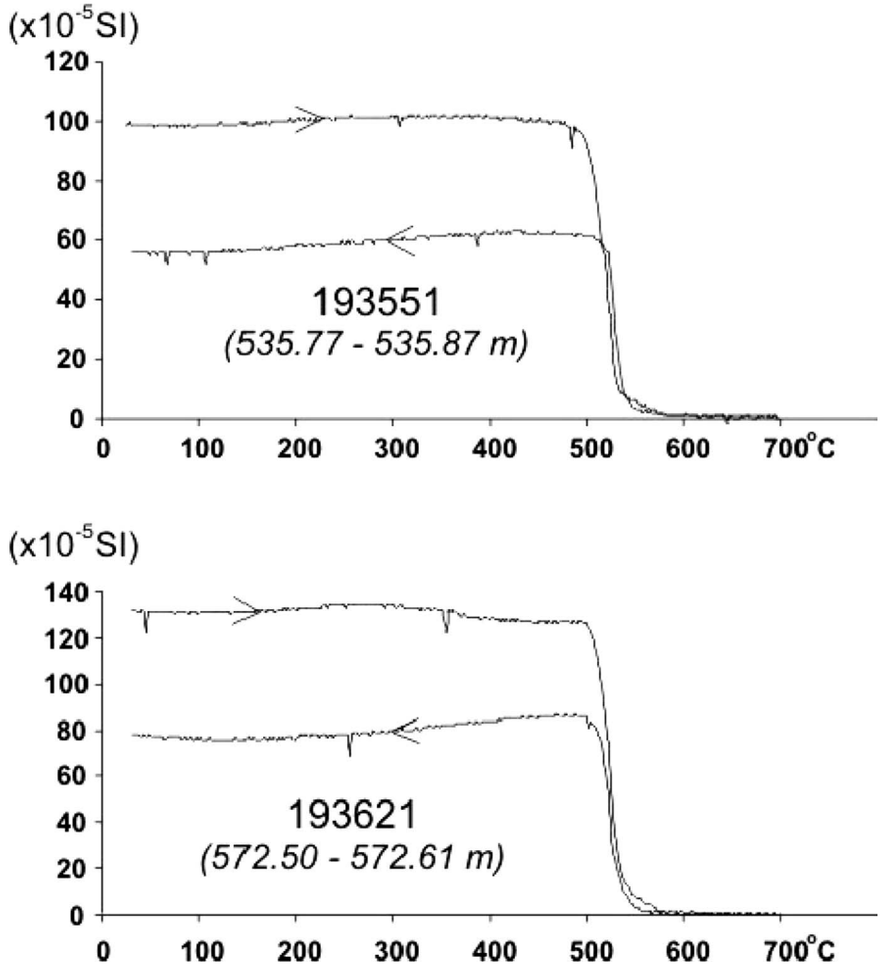


Figure 6 Analysis of Curie points using magnetic susceptibility *vs* temperature curves for two specimens of Keene Basalt. Measurements of susceptibility were made at the corresponding temperature.

Kanpa Formation

Nine samples of red mudstone and sandstone from the base of the Kanpa Formation were collected from two oriented intervals (639.50–688.82 m and 702.41–706.37 m). The orientation of the deeper interval was verified by independent features at each end. Six samples yielded internally consistent, stable southwards-down remanence directions with unblocking temperatures between 500 and 630°C (Figures 3d, 4d). Magnetic susceptibility did not change significantly during thermal treatment, except for a minor increase between 500 and 600°C (Figure 5a) that was not associated with any change in remanence direction (Figure 3d). This part of the drillcore appears very suitable for palaeomagnetic study, and data from additional samples would improve the reliability of the result (Table 1). Oriented samples from the top part of the Kanpa Formation (519.40–522.17 m) were magnetically unstable.

Keene Basalt

This core interval contains numerous fractures in different orientations, making their signatures difficult to identify in the acoustic scanner composite log. Twelve samples were measured from four oriented intervals between 531.17 and 574.94 m. All samples yielded identical single-component stable magnetisations that have high coercivities and unblocking temperatures distributed between 570 and 700°C (Figure 3e). Thermal and AF demagnetisation produced similar results. Remanence declinations are similar within each oriented interval, but vary between oriented intervals (Figure 4e), indicating that at least three (possibly all four) oriented intervals were incorrectly oriented. The inconsistent azimuthal orientations could be the result of the highly magnetic basalts having affected the magnetic compass in the acoustic scanner. The inclinations of all oriented intervals are tightly grouped, with a mean of 32°, corresponding to a palaeolatitude of 17.6° (Table 1). Only a single magnetic polarity is present. Analysis of Curie temperatures (Figure 6) indicates the dominance of almost pure magnetite, although thermal demagnetisation curves (Figure 3e) show that hematite is also present. Some magnetite was oxidised to hematite during heating to temperatures above 500°C, as indicated by decreases in magnetic susceptibility (Figures 5, 6).

Wahlgu Formation

Thirteen samples of red-brown diamictite were obtained from five oriented intervals close to the base of the Wahlgu Formation (428.99–461.49 m).

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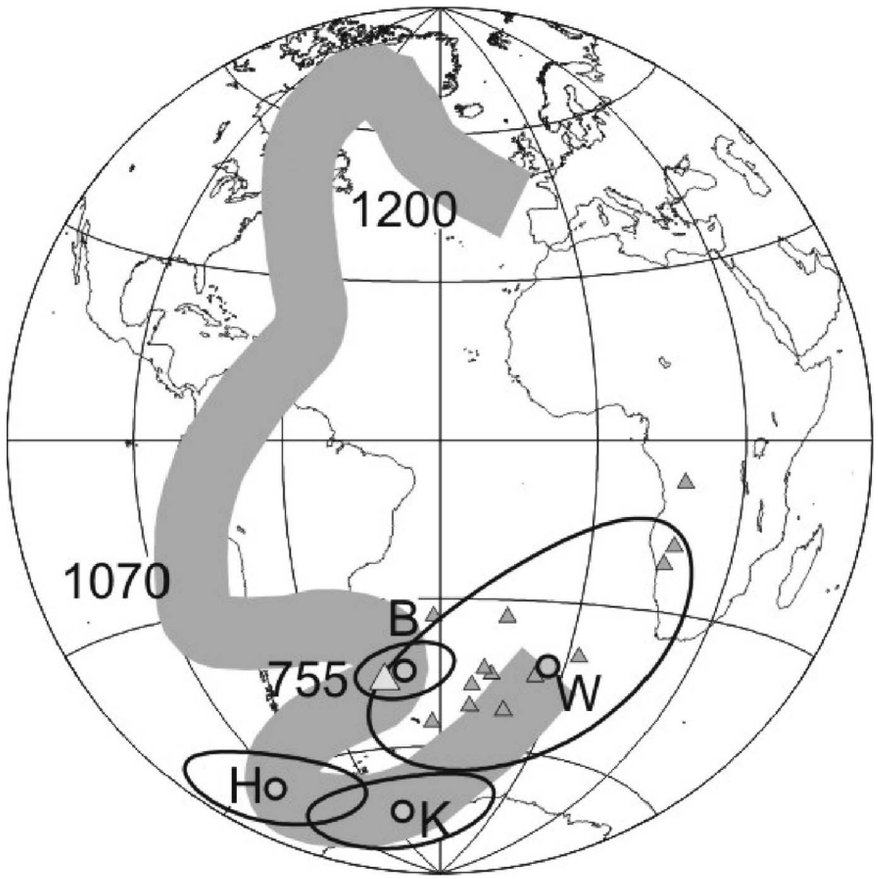


Figure 7 Mesoproterozoic to latest Neoproterozoic palaeopoles for Australia. Open dots with 95% confidence ovals indicate poles from this study: W, Wahlgu; K, Kanpa; H, Hussar; B, Browne (see Table 1). Triangular symbols represent other Australian poles between 755 and 530 Ma; the large triangle represents the result for the 755 Ma Mundine Well dykes (Wingate & Giddings 2000). The shaded path is a revised apparent polar wander path for Australia between 1200 and 600 Ma (Pisarevsky *et al.* 2003a, and this study).

Thermal demagnetisation isolated a highly dispersed low-temperature component and a medium- to high-temperature component directed shallowly to the southwest. An example of thermal demagnetisation is shown in Figure 3f; unblocking temperatures are mostly above 600°C, indicating that hematite is a main remanence carrier. The high-temperature component is also dispersed (Figure 4f) both within the core and between cores, which is typical of diamictite. The absence of correlation between oriented intervals and palaeomagnetic direction (Figure 4f) indicates that core orientation is correct.

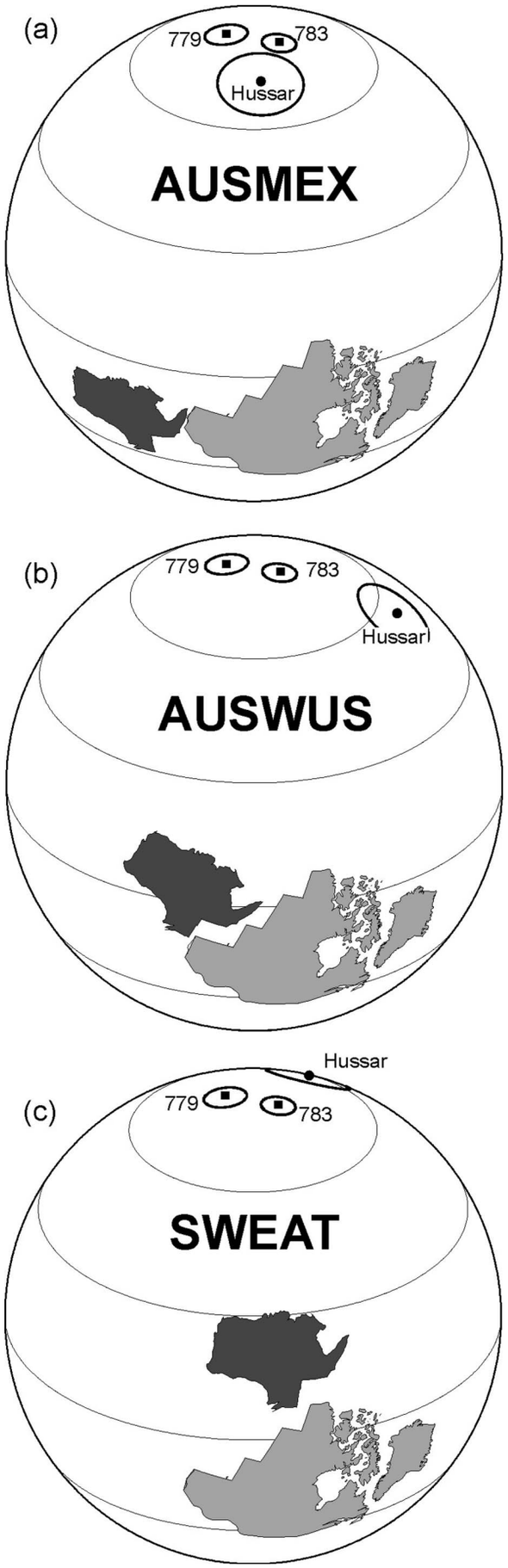
DISCUSSION

Thermal demagnetisation of red mudstone, siltstone and sandstone of the *ca* 830–800 Ma Browne Formation isolated a stable, high-temperature, unipolar remanence component carried by hematite. Its mean inclination of −33.8° corresponds to low to intermediate palaeolatitudes (−18.5°). The magnetisation is likely to be primary because there is abundant evidence for dry and warm environments during deposition of the Browne Formation (e.g. the presence of salt), which indicates a depositional to early diagenetic age for the hematite. Moreover, the palaeomagnetic and rock-magnetic characteristics of Browne Formation samples in Lancer 1 are very similar to those reported for samples of Browne Formation in Empress 1A, located 260 km southeast of Lancer 1. The Browne Formation palaeopole is close to the primary pole for the 755 Ma Mundine Well dykes (Figure 7). If the age of the Browne Formation is correct, this implies that little motion of the Australian Plate occurred between *ca* 830 and 755 Ma. Alternatively, the age of the Browne Formation may be slightly younger than 830–800 Ma.

Red mudstones of the *ca* 800–760 Ma Hussar and *ca* 760–720 Ma Kanpa Formations also yielded stable, if slightly dispersed, remanence directions. The mean inclinations of +4.6° for the Hussar Formation and +17.7° for the Kanpa Formation indicate that the Australian Plate traversed the equator some time between deposition of the Browne and Kanpa Formations. The presence of two polarities is a point in favour of a primary magnetisation and suggests that magnetostratigraphic correlation may be possible with strata in other drillholes and other basins (e.g. Georgina and Amadeus Basins, and Adelaide Rift Complex). The position of the 755 Ma Mundine pole at some distance from the Hussar and Kanpa poles, but close to the Browne pole, suggests that the precise ages of these three formations may be slightly different from the estimations shown in Table 1. However, it does not contradict the broad correlation with the Buldya Group (Grey *et al.* 2005).

Palaeopoles for the Browne and Hussar Formations are distinct from latest Neoproterozoic and Phanerozoic Australian palaeopoles (McElhinny & McFadden 2000 table 6.14; McElhinny *et al.* 2003 table 1), which also

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argues in favour of primary magnetisations. The palaeopole for the Kanpa Formation is close to some Middle Devonian Australian poles. However, this pole is based on only six samples and should be considered as a preliminary result.

Although the azimuthal orientations of Keene Basalt samples could not be established, consistent palaeolatitudes (mean = 17.6°) were determined from reliable inclination measurements. The inconsistent azimuthal orientations measured in the Keene Basalt may be the result of the compass in the acoustic scanner being affected by highly magnetic minerals in the basalt.

Palaeomagnetic directions in diamictite of the *ca* 600 Ma Wahlgu Formation are dispersed and yield an imprecise palaeopole that corresponds broadly to those observed from other latest Neoproterozoic poles for Australia (Figure 7). The low palaeolatitude of $7.3 \pm 16.5°$ confirms those measured in the Wahlgu Formation in Empress 1A (Pisarevsky *et al.* 2001) and is consistent with correlation of the Wahlgu Formation with the Marinoan glaciogenic Elatina Formation in the Adelaide Rift Complex (Grey *et al.* 2005). Low latitudes during deposition of Neoproterozoic glacial strata remain enigmatic.

All inclination data obtained from Lancer 1 indicate that Australia occupied low to moderate latitudes during Middle and Late Neoproterozoic time. This confirms similar conclusions from other palaeomagnetic studies, including results from the Empress 1A and Vines 1 drillholes (Pisarevsky 2001; Pisarevsky *et al.* 2001). New data from this study also imply relatively slow movement of the Australian Plate during the Late Neoproterozoic.

Palaeopoles from this study are illustrated in Figure 7, together with Late Neoproterozoic to Cambrian palaeopoles for Australia extracted from the Global Palaeomagnetic Database (Pisarevsky 2005). The previous data include a high-quality pole for the 755 Ma Mundine Well dykes (Wingate & Giddings 2000) and a revised apparent polar wander path for Australia (Pisarevsky *et al.* 2003a) that incorporates poles for the 1075 Ma Warakurna Large Igneous Province (Wingate 2002, 2003; Wingate *et al.* 2002, 2004) and the 1212 Ma Fraser dykes (Pisarevsky *et al.* 2003a). The large 95% confidence ovals (which can be improved by additional sampling) and the lack of other Australian palaeopoles between 1070 and 755 Ma make it difficult to use

Figure 8 Australia–Laurentia reconstructions and their palaeomagnetic constraints at *ca* 780 Ma, based on comparison of the Hussar Formation pole (this study) with coeval Laurentian data. (a) AUSMEX (Wingate *et al.* 2002); (b) AUSWUS (Brookfield 1993; Karlstrom *et al.* 1999; Burrett & Berry 2000); (c) SWEAT (Dalziel 1991; Hoffman 1991; Moores 1991). Rotation parameters (Euler pole latitude, longitude, rotation angle, negative for clockwise): Laurentia to absolute framework (a–c) 19.7, −153.1, −87.9; Australia to Laurentia (a) 54.5, 149.9, 120.2; (b) 49.9, 115.7, 114.4; (c) 32.5, 122.4, 130.7. Squares, Laurentian poles: 779 Ma Tsezotene sills and dykes (Park *et al.* 1989; LeCheminant & Heaman 1994) and 785–782 Ma Wyoming Dykes (Harlan *et al.* 1997).

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palaeopole positions to date the studied rocks precisely. The sequence of palaeopoles for the Browne, Hussar and Kanpa Formations defines an apparent polar wander path for Australia between *ca* 820 and *ca* 720 Ma (Figure 7).

The Neoproterozoic palaeomagnetic database for Laurentia (ancestral North America) is also poor (see Pisarevsky *et al*. 2003b for an overview), although there are relatively reliable palaeopoles for the 779 Ma Tsezotene intrusions (Park *et al*. 1989; LeCheminant & Heaman 1994) and the 785–782 Ma Wyoming dykes (Harlan *et al*. 1997), which represent the latest pre-breakup, or earliest breakup, period of Rodinia's history. These poles are broadly coeval with the Hussar Formation, and so they can be used to test reconstructions between Australia and Laurentia. The palaeopole from the Hussar Formation in Lancer 1 is calculated from palaeomagnetic directions of 17 samples, and its reliability is supported by the presence of two polarities. Slightly better correspondence of Laurentian poles with the Hussar Formation pole is achieved for the AUSMEX reconstruction, compared to the SWEAT or AUSWUS models (Figure 8), implying that if Australia and Laurentia were joined within Rodinia, the AUSMEX configuration is the most likely.

Palaeomagnetic data from cores not oriented in azimuth of Browne and Hussar Formations in the Empress 1A drillhole (Pisarevsky *et al*. 2001) did not reveal any useful magnetostratigraphic pattern because the remanence inclination is shallow, and no declination data were available. However, a dipmeter log is available in Empress 1A, and independent azimuthal orientation of samples in this drillhole is possible in future studies. In Lancer 1, the studied oriented intervals in the Browne Formation are of a single polarity (Figure 4b), and there are two polarities in the Hussar Formation (Figure 4c). Additional sampling of these two formations in Lancer 1 and Empress 1A should be undertaken to build a magnetostratigraphic column, which would be a very useful correlation tool for Neoproterozoic sedimentary successions throughout Australia and globally.

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REFERENCES

Apak S. N., Moors H. T. & Stevens M. K. (Compilers) 2002. GSWA Vines 1 well completion report, Waigen area, Officer Basin, Western Australia. *Geological Survey of Western Australia Record* **2001/18**.

Brookfield M. E. 1993. Neoproterozoic Laurentia–Australia fit. *Geology* **21**, 683–686.

Burrett C. & Berry R. 2000. Proterozoic Australia–Western United States (AUSWUS) fit between Laurentia and Australia. *Geology* **28**, 103–106.

Carlsen G. M. & Grey K. 1998. GSWA Empress 1A and the petroleum potential of the Neoproterozoic Officer Basin. *Geological Survey of Western Australia Annual Review 1997–98*, 119–126.

Cottrell R. D. & Tarduno, J. A. 2003. A Late Cretaceous pole for the Pacific plate: implications for apparent and true polar wander and the drift of hotspots. *Tectonophysics* **362**, 321–333.

Dalziel I. W. D. 1991. Pacific margins of Laurentia and East Antarctica–Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. *Geology* **19**, 598–601.

Dalziel I. W. D. 1997. Neoproterozoic–Paleozoic geography and tectonics: review, hypothesis, environmental speculation. *Geological Society of America Bulletin* **109**, 16–42.

Fisher R. A. 1953. Dispersion on a sphere. *Proceedings of the Royal Society of London* **A217**, 295–305.

Grey K., Hocking R. M., Stevens M. K., Bagas L., Carlsen G. M., Irimies F., Pirajno F., Haines P. W. & Apak S. N. 2005. Lithostratigraphic nomenclature of the Officer Basin and correlative parts of the Paterson Orogen, Western Australia. *Geological Survey of Western Australia Report* **93**.

Haines P. W., Mory A. J., Stevens M. K. & Ghori K. A. R. 2004. GSWA Lancer 1 well completion report (basic data), Officer and Gunbarrel Basins, Western Australia. *Geological Survey of Western Australia Record* **2004/10**.

Harlan S. S., Geissman J. W. & Snee L. W. 1997. Paleomagnetic and $^{40}Ar/^{39}Ar$ geochronologic data from late Proterozoic mafic dykes and sills, Montana and Wyoming. *US Geological Survey Professional Paper* **1580**, 1–16.

Hoffman P. F. 1991. Did the breakout of Laurentia turn Gondwana inside out? *Science* **252**, 1409–1412.

Hoffman P. F., Hawkins D. P., Isachsen C. E. & Bowring S. A. 1996. Precise U–Pb zircon ages for early Damaran magmatism in the Summas Mountains and Welwitschia Inlier, northern Damara belt, Namibia. *Communications of the Geological Survey of Namibia* **11**, 47–52.

Karlstrom K. E., Harlan S. S., Williams M. L., McLelland J., Geissman J. W. & Åhall K-I. 1999. Refining Rodinia: geologic evidence for the Australia–Western U.S. connection in the Proterozoic. *GSA Today* **9**(10), 1–7.

Kirschvink J. L. 1980. The least squares line and plane and the analysis of palaeomagnetic data. *Geophysical Journal of the Royal Astronomical Society* **62**, 699–718.

LeCheminant A. N. & Heaman L. M. 1994. 779 Ma mafic magmatism in the northwestern Canadian Shield and northern Cordillera: a new regional time-marker. *In*: *8th International Conference. Geochronology, Cosmochronology and Isotope Geology, Program Abstracts (Berkeley, California)*, p. 197. US Geological Survey Circular **1107**.

McElhinny M. W. & McFadden P. L. 2000. *Paleomagnetism: Continents and Oceans*. Academic Press, San Diego.

McElhinny M. W., Powell C. McA. & Pisarevsky S. A. 2003. Paleozoic terranes of Eastern Australia and the drift history of Gondwana. *Tectonophysics* **362**, 41–65.

Moores E. M. 1991. Southwest US–East Antarctic (SWEAT) connection: a hypothesis. *Geology* **19**, 425–428.

Mory A. J. & Haines P. W. (Editors) 2005. GSWA Lancer 1 well completion report (interpretative papers), Officer and Gunbarrel Basins, Western Australia. *Geological Survey of Western Australia Record* **2005/4**.

Park J. K., Norris D. K. & Larochelle A. 1989. Paleomagnetism and the origin of the Mackenzie Arc of northwestern Canada. *Canadian Journal of Earth Sciences* **26**, 2194–2203.

Pisarevsky S. A. 2001. Paleomagnetic study of Vines 1. *Geological Survey of Western Australia Record* **2001/18**, Appendix 4.

Pisarevsky S. A. 2005. New edition of the Global Paleomagnetic Database. *EOS* **86**, 170.

Pisarevsky S. A., Li Z. X., Grey K. & Stevens M. K. 2001. A paleomagnetic study of the Empress 1A corehole in the Officer Basin: no evidence for a high-latitude position for Australia in the Neoproterozoic. *Precambrian Research* **110**, 93–108.

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PISAREVSKY S. A., WINGATE M. T. D. & HARRIS L. B. 2003a. Late Mesoproterozoic (*ca* 1.2 Ga) paleomagnetism of the Albany–Fraser orogen: no pre-Rodinia Australia–Laurentia connection. *Geophysical Journal International* **155**, F6–F11.

PISAREVSKY S. A., WINGATE M. T. D., POWELL C. MCA., JOHNSON S., & EVANS D. A. D. 2003b. Models of Rodinia assembly and fragmentation. *In*: Yoshida M., Windley B. & Dasgupta S. eds. *Proterozoic East Gondwana: Supercontinent Assembly and Breakup*, pp. 35–55. Geological Society of London Special Publication **206**.

PREISS W. V. 2000. The Adelaide Geosyncline of South Australia and its significance in Neoproterozoic continental reconstruction. *Precambrian Research* **100**, 21–63.

SIMEONOVA A. P. & IASKY R. P. 2005. Seismic mapping, salt deformation, and hydrocarbon potential of the central western Officer Basin, Western Australia. *Geological Survey of Western Australia Report* **96**.

STEVENS M. K. & ADAMIDES N. G. 1998. GSWA Trainor 1 well completion report, Savory Sub-basin, Western Australia, with notes on petroleum and mineral potential. *Geological Survey of Western Australia Record* **1996/12**.

STEVENS M. K. & APAK S. N. 1999. Empress 1 and 1A well completion report, Yowalga Sub-basin, Officer Basin, Western Australia. *Geological Survey of Western Australia Record* **1999/4**.

WALTER M. R., VEEVERS J. J., CALVER C. R. & GREY K. 1995. Neoproterozoic stratigraphy of the Centralian Superbasin, Australia. *Precambrian Research* **73**, 173–195.

WEIL A. B., VAN DER VOO R., MAC NIOCAILL C. & MEERT J. G. 1998. The Proterozoic supercontinent Rodinia: paleomagnetically derived reconstruction for 1100 to 800 Ma. *Earth and Planetary Science Letters* **154**, 13–24.

WINGATE M. T. D. 2002. Age and paleomagnetism of Mesoproterozoic dolerite sills of the western Bangemall Basin, Western Australia. *Geological Survey of Western Australia Record* **2002/4**.

WINGATE M. T. D. 2003. Age and paleomagnetism of dolerite intrusions of the southeastern Collier Basin, and the Earaheedy and Yerrida Basins, Western Australia. *Geological Survey of Western Australia Record* **2003/3**.

WINGATE M. T. D. & GIDDINGS J. W. 2000. Age and paleomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia–Laurentia connection at 755 Ma. *Precambrian Research* **100**, 335–357.

WINGATE M. T. D., PIRAJNO F. & MORRIS P. A. 2004. The Warakurna large igneous province: a new Mesoproterozoic large igneous province in west-central Australia. *Geology* **32**, 105–108.

WINGATE M. T. D., PISAREVSKY S. A. & EVANS D. A. D. 2002. Rodinia connections between Australia and Laurentia: no SWEAT, no AUSWUS? *Terra Nova* **14**, 121–128.

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